Aga Khan University Examination Board

Notes from E-Marking Centre on SSC-II Mathematics Examination April/ May 2019

Introduction

This document has been produced for the teachers and candidates of Secondary School Certificate (SSC) part II Mathematics. It contains comments on candidates' responses to the 2019 SSC-II Examination, indicating the quality of the responses and highlighting their relative strengths and weaknesses.

E-Marking Notes

This includes overall comments on candidates' performance on every question and some specific examples of candidates' responses which support the mentioned comments. Please note that the descriptive comments represent an overall perception of the better and weaker responses as gathered from the e-marking session. However, the candidates' responses shared in this document represent some specific example(s) of the mentioned comments.

Teachers and candidates should be aware that examiners may ask questions that address the Student Learning Outcomes (SLOs) in a manner that require candidates to respond by integrating knowledge, understanding and application skills they have developed during the course of study. Candidates are advised to read and comprehend each question carefully before writing the response to fulfil the demand of the question.

Candidates need to be aware that the marks allocated to the questions are related to the answer space provided on the examination paper as a guide to the length of the required response. A longer response will not in itself lead to higher marks. Candidates need to be familiar with the command words in the SLOs which contain terms commonly used in examination questions. However, candidates should also be aware that not all questions will start with or contain one of the command words. Words such as 'how', 'why' or 'what' may also be used.

General Observations

In weaker responses, it was noted that the candidates were not well versed with the equations involving radicals, simultaneous equations using graphical method and the topic related to radian measure. They performed well in applying distance and midpoint formula. It was seen that candidates responded well in applying theorems related to chords of a circle.

Detailed Comments:

Constructed Response Questions (CRQs)

Question 1:

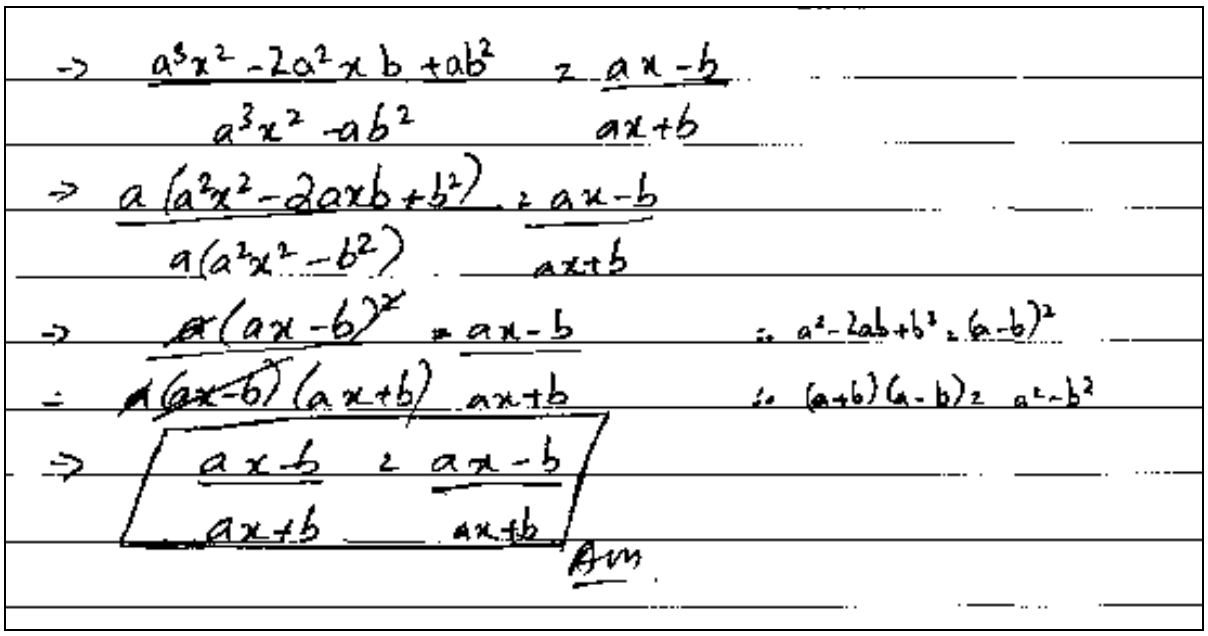
This question offered a choice between part **a** and part **b**. Candidates chose to attempt part **a** more than part **b**.

Question 1a:

Simplify $\left(a^3x^2 - 2a^2xb + ab^2\right) \div \left(a^3x^2 - ab^2\right)$ to express as $\frac{ax-b}{ax+b}$.

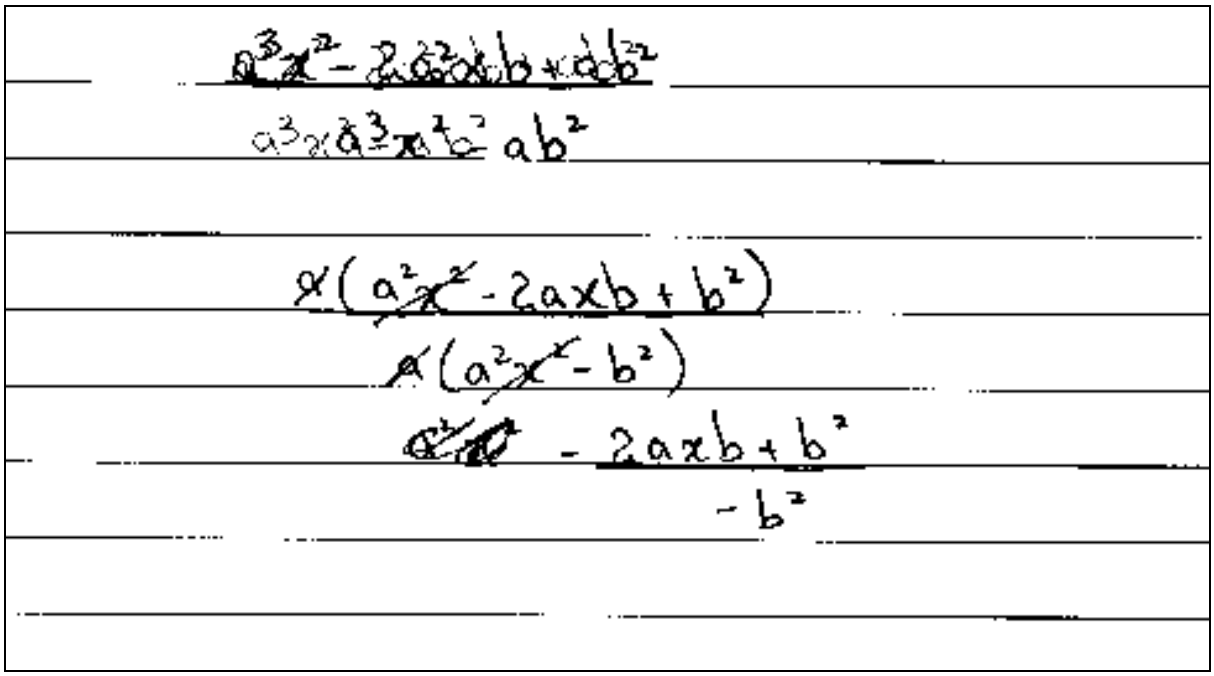
Better responses indicated that candidates had good command over the concept simplifying the rational expression. They started well by taking a as common from $a^3x^2 - 2a^2xb + ab^2$ and $\left(a^3x^2 - ab^2\right)$ and consequently applied the formula of $(a-b)^2$ and $a^2 - b^2$. Finally, they were able to simplify the given rational expression and reached the required form.

Example:



Weaker responses showed that candidates had lack of understanding in simplifying the rational expression. They made mistakes in applying the formula of $(a-b)^2$ and a^2-b^2 and as a result, got confused to simplify the given algebraic expressions.

Example:



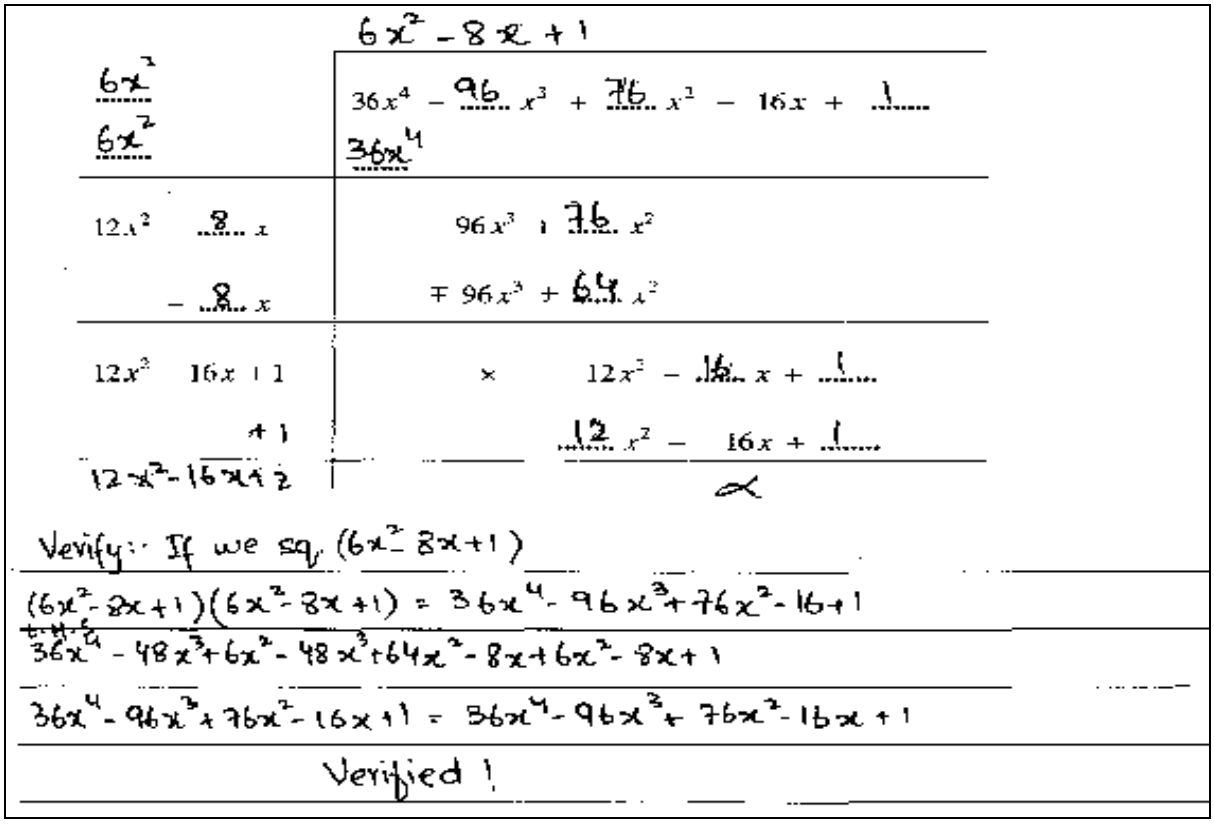
Question 1b:

Complete the solution of the square root of an algebraic expression by filling the space provided. Hence, verify your answer.

....	$36x^4 - \ldots\ldots x^3 + \ldots\ldots x^2 - 16x + \ldots\ldots$ .....
$12x^2 - \ldots\ldots x$ $- \ldots\ldots x$	$- 96x^3 + \ldots\ldots x^2$ $\mp 96x^3 \pm \ldots\ldots x^2$
$12x^2 - 16x + 1$	$\times \quad 12x^2 - \ldots\ldots x + \ldots\ldots$ $\ldots\ldots x^2 - 16x + \ldots\ldots$

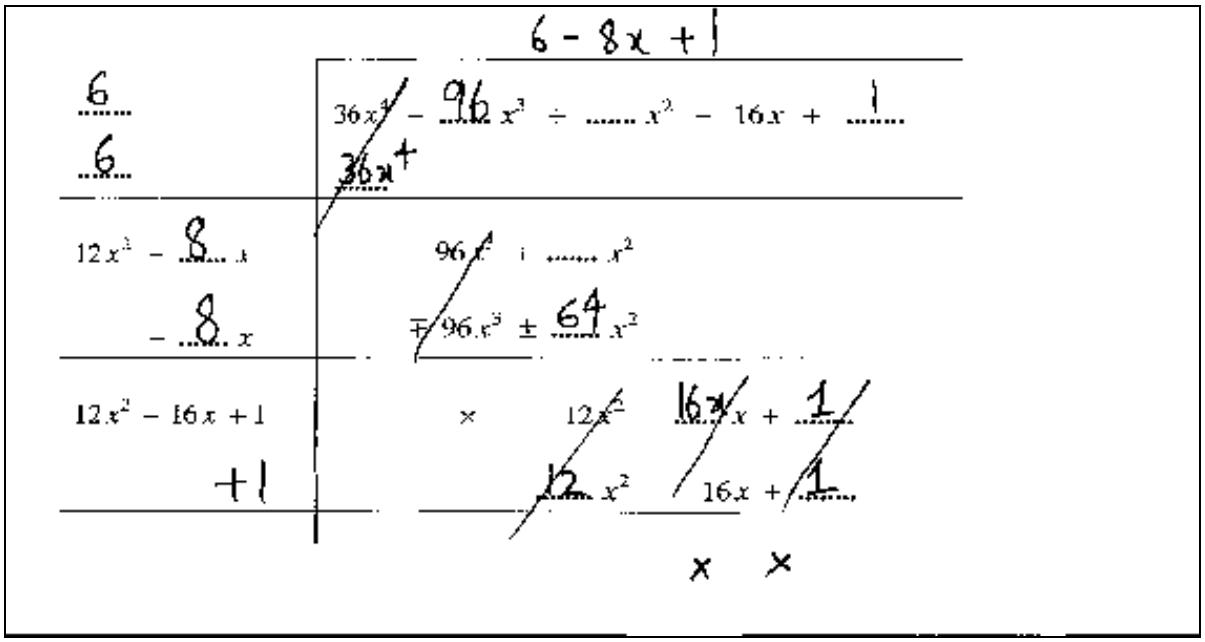
Better responses displayed that candidates comprehended the question well and applied the correct process of division method to find the square root of the given algebraic expression and verified the answer accordingly.

Example:



Weaker responses reflected that candidates made mistakes in the division process. They failed to follow the steps needed to find the square root of the given algebraic expression. The mistakes are noted in writing divisor, quotient and sign in the process of addition and subtraction of algebraic terms.

Example 1:

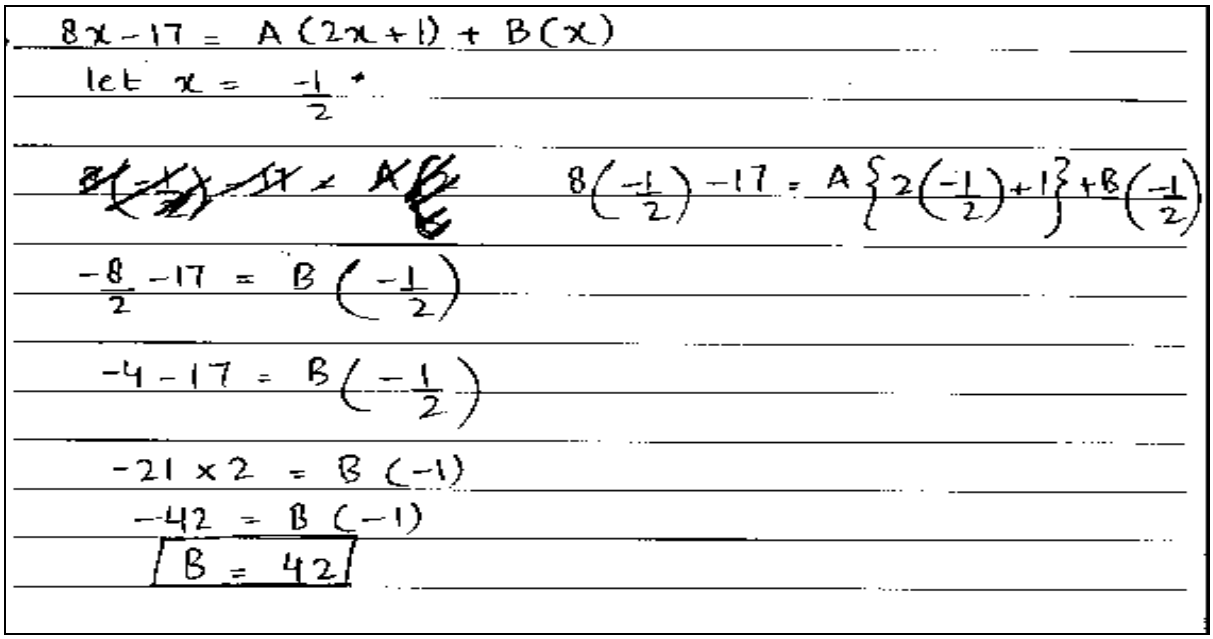


Question 2:

If $\frac{8x-17}{x(2x+1)} = \frac{A}{x} + \frac{B}{2x+1}$, then find the value of *B*.

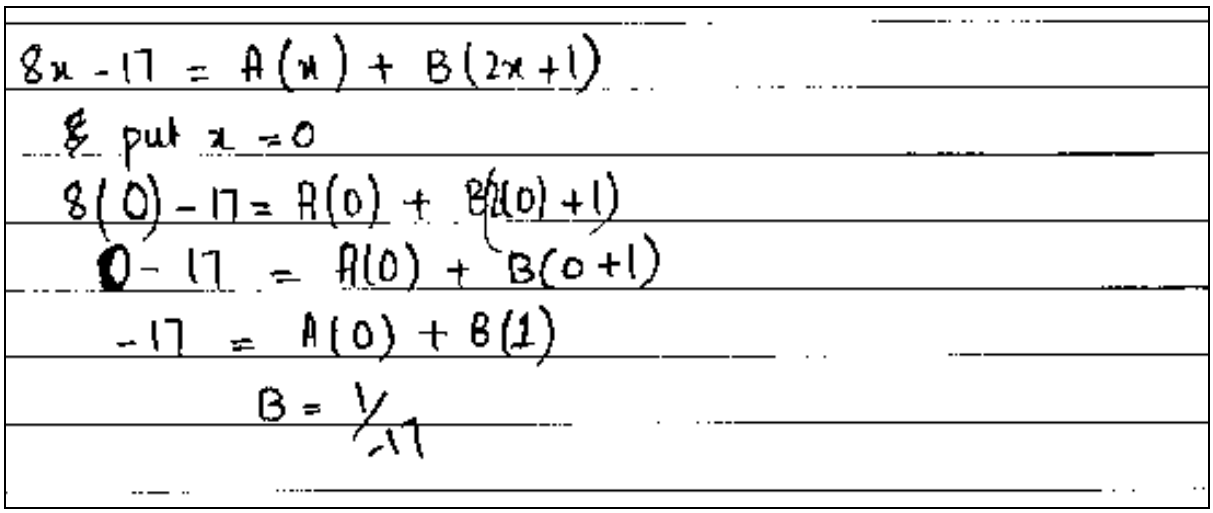
Better responses depicted that candidates started well by taking LCM and then substituted $x = -\frac{1}{2}$ in $A(2x+1)+Bx = 8x-17$. This helped candidates to get the value of *B* quite easily.

Example:



Weaker responses revealed that candidates failed in taking LCM and got stuck in further steps. Because of this, they could not reach the desired result.

Example:



Question 3:

This question offered a choice between part **a** and part **b**. Candidates chose to attempt part **b** more than part **a**.

Question 3a:

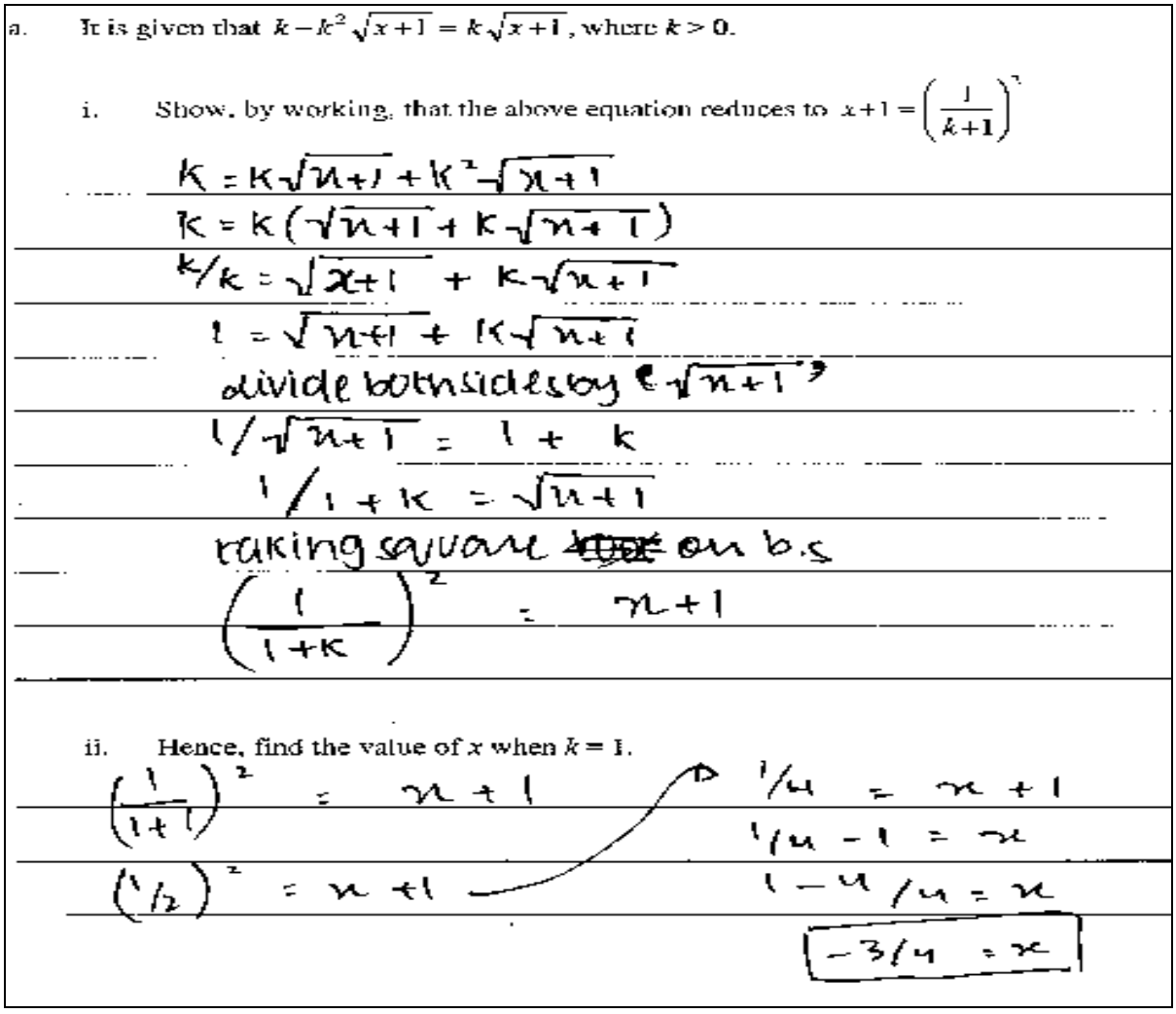
It is given that $k-k^2\sqrt{x+1}=k\sqrt{x+1}$, where $k > 0$.

i. Show, by working, that the above equation reduces to $x+1=\left(\frac{1}{k+1}\right)^2$

ii. Hence, find the value of x when $k = 1$.

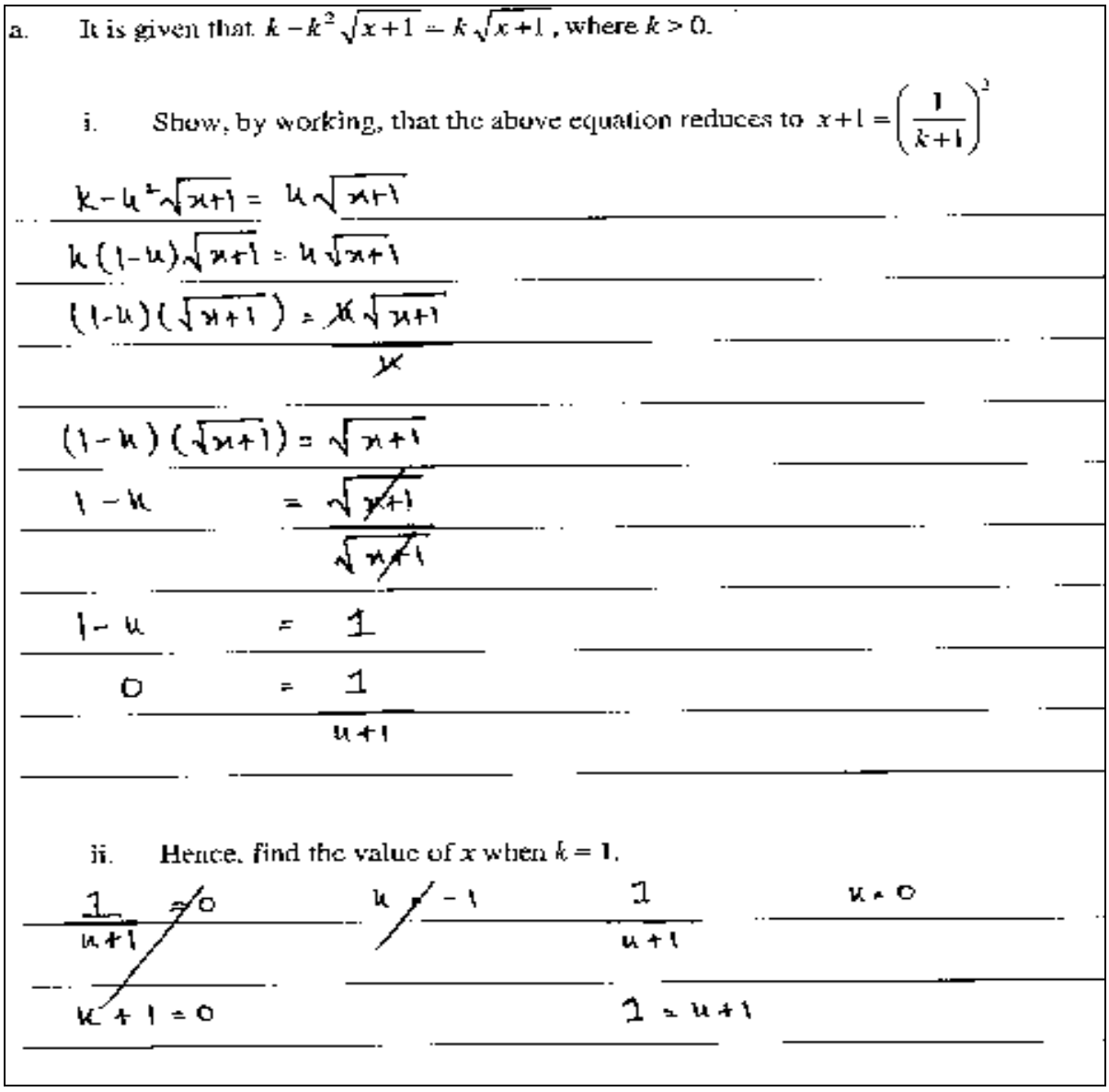
Better responses displayed that candidates systematically solved the given linear equation by using correct mathematical steps. They started well by taking k as common and cancelled from both the sides. After correct simplification and squaring on both sides, they were able to manage the required form. Hence, they got the desired value of x as well.

Example 1:



Weaker responses exhibited that candidates failed to solve the given linear equation. It was also seen that the candidates did not understand the question in order to convert the required form and hence failed to perform the algebraic operation correctly on the given equation. They were unable to apply the steps to solve the equation involving square roots or radical sign to get the required form.

Example:



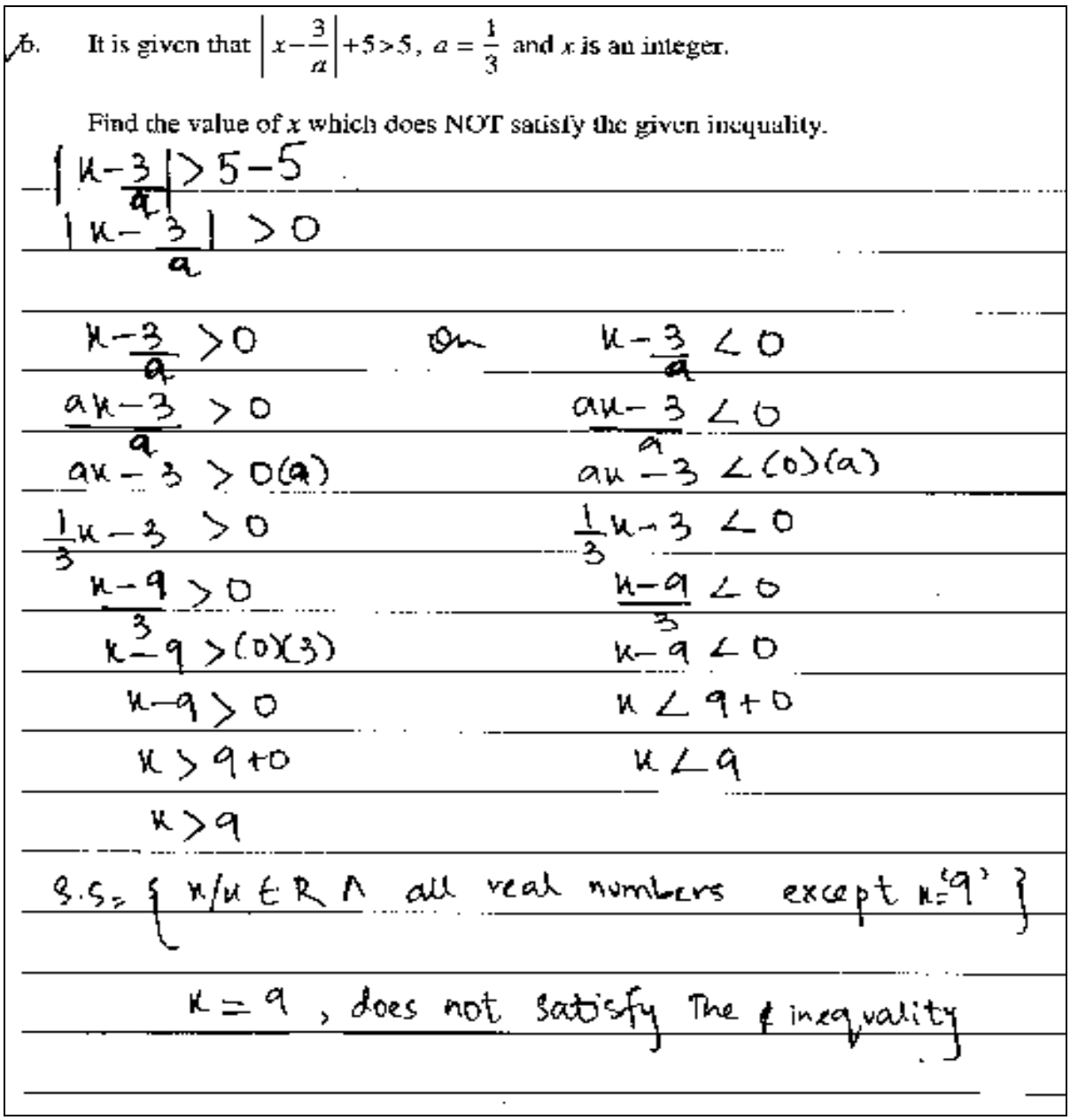
Question 3b:

It is given that $\left|x-\frac{3}{a}\right|+5>5$, $a=\frac{1}{3}$ and x is an integer.

Find the value of x which does NOT satisfy the given inequality.

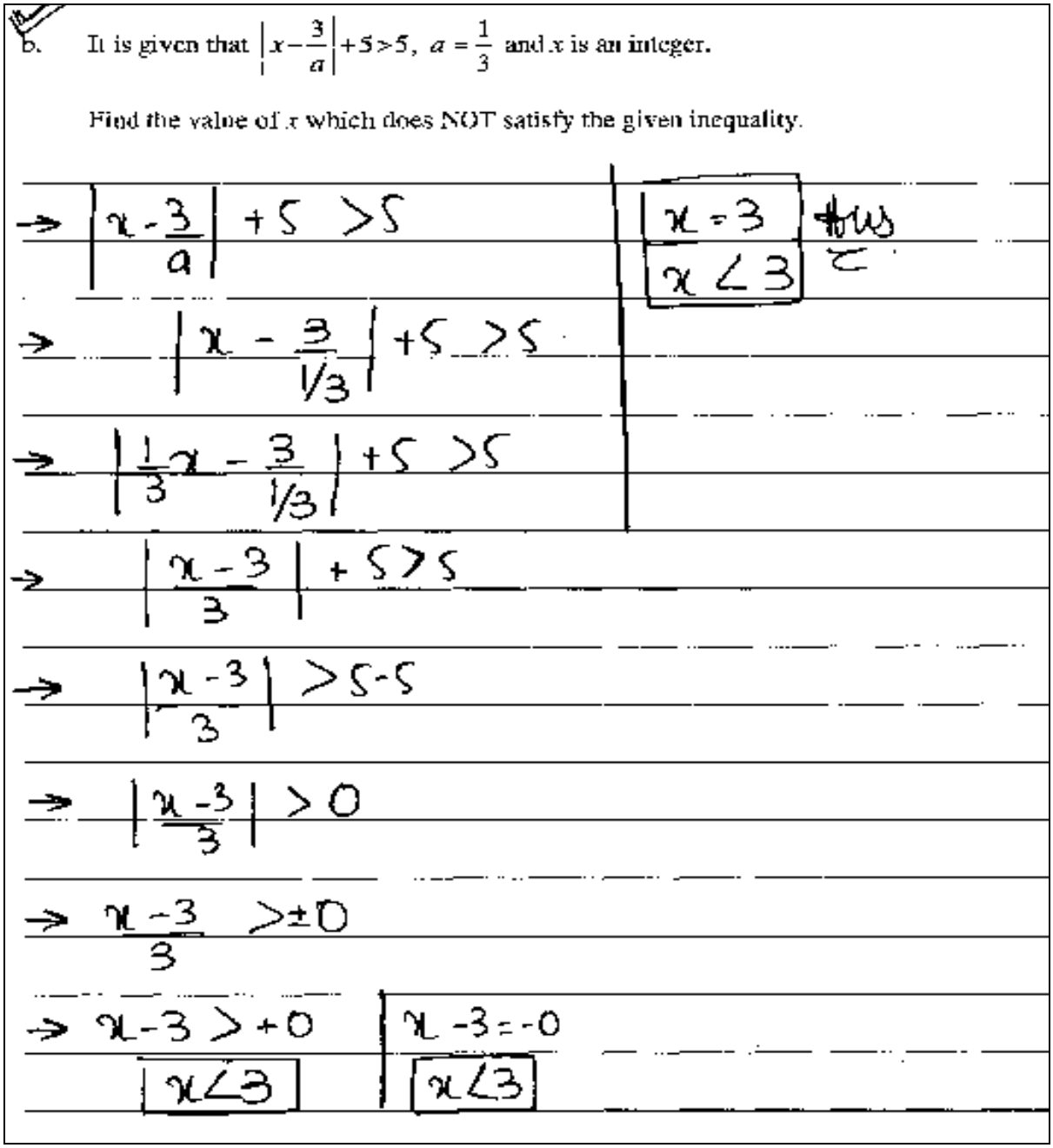
Better responses demonstrated that candidates were able to solve the inequality and wrote the desired answer to fulfill the requirement of the question.

Example:



Weaker responses revealed that candidates were confused in solving equation involving radical sign and inequalities. The weaker responses showed mistakes in performing algebraic operations. It was also seen that candidates were unable to get the value of x that does not satisfy the given inequality.

Example:



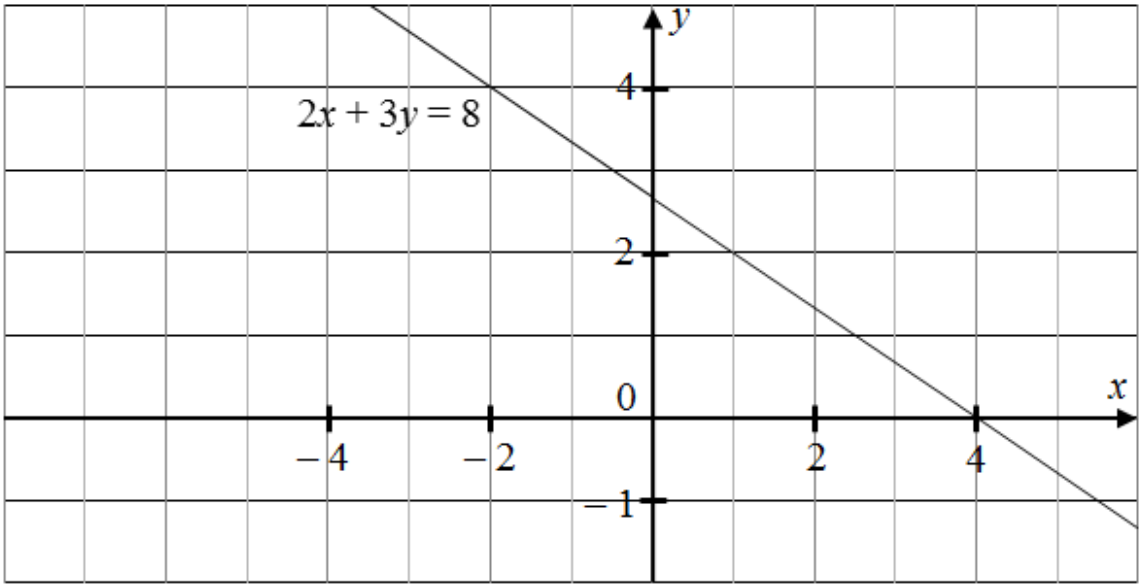
Question 4:

Given that $\left\{\left(1, \frac{a}{2}\right)\right\}$ is the solution set of simultaneous linear equations $x+y=3$ and $2x+3y=8$.

a. Line $2x+3y=8$ is shown in the given graph. On the same graph, draw another line $x+y=3$ by taking two points only.

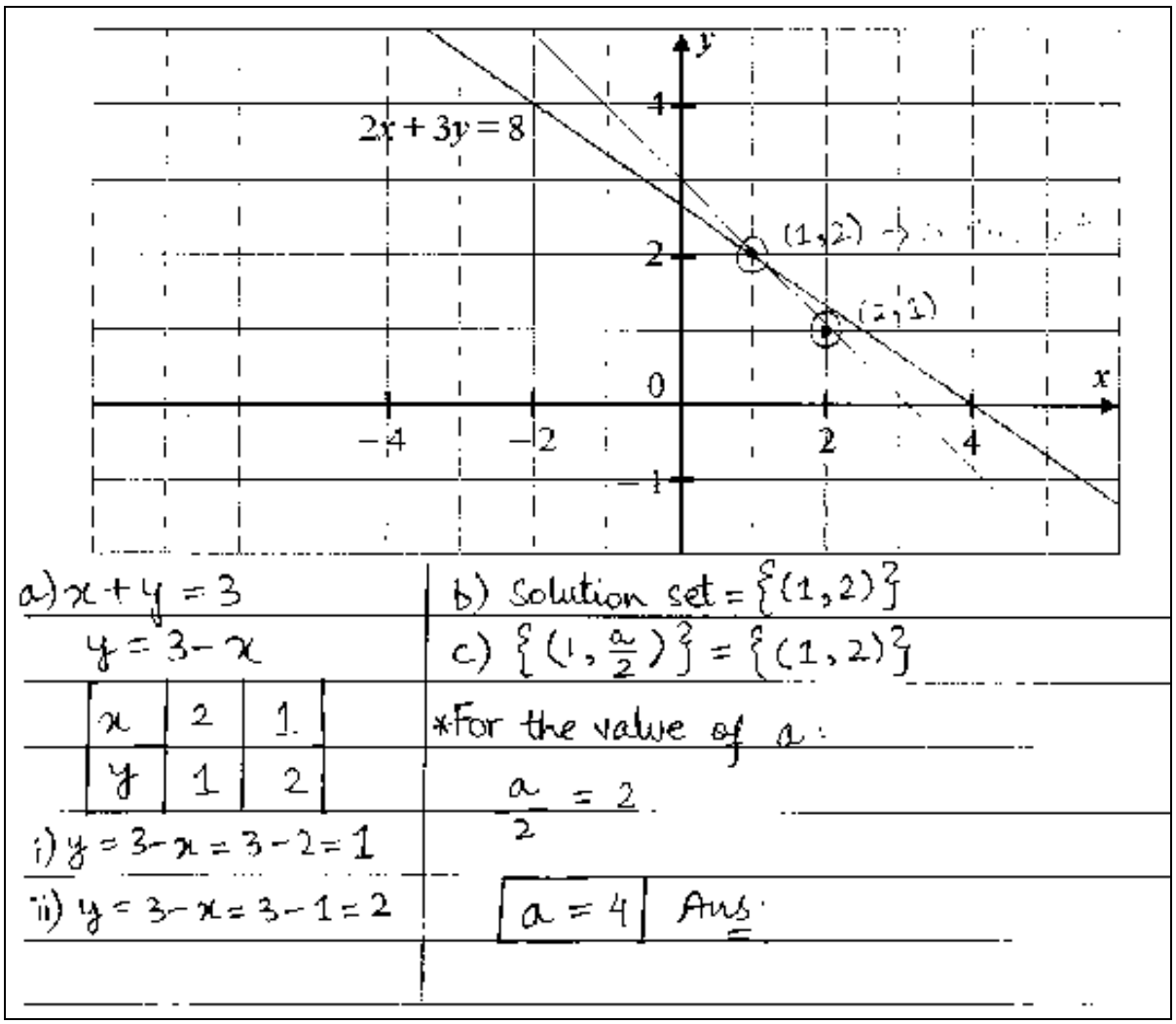
b. From the graph, find the solution set.

c. Hence, find the value of *a*.



Better responses displayed that candidates proficiently drew the line by taking the values of *x* and *y*. This helped candidates to find the solution set from graph and hence they were able to find the correct value of *a*.

Example:

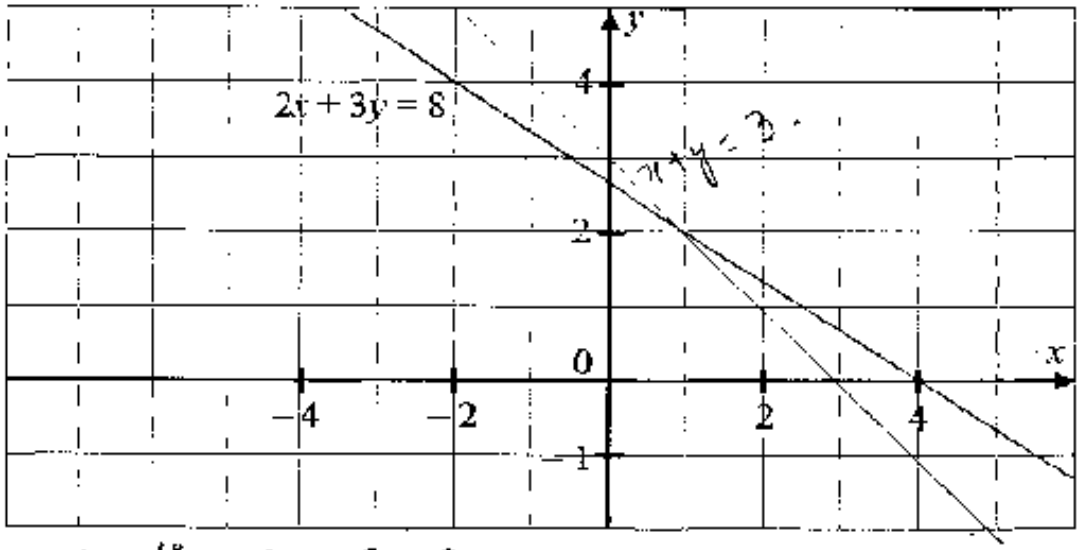


Weaker responses exposed various types of mistakes. Some common mistakes have been listed below.

- Candidates failed to find the unknown values of *x* and *y*, which is indicative of the fact that the candidates are weak in applying basic arithmetic operations to solve linear equation in two variables.
- Candidates failed to take the proper scale on x-axis or y-axis
- Candidates showed misconceptions in locating points on the graph papers. They interchanged the values of *x* and *y* and failed to differentiate between − and + sign while locating the points on the graph paper.

It was also seen that fewer candidates found the solution set by elimination or substitution method rather than from the given graph. Although it was clearly mentioned in the question that solution set need to be calculated from graph and therefore candidates lost their marks.

Example:



b. Solution Set = {2}

- Value of a:

$\{(1, \frac{7}{2})\}$ $\{(1, \frac{9}{2})\} \Rightarrow 2(1) = a$

$2 = a$.

So, $(1, \frac{7}{2}) \Rightarrow \{(1, 1)\}$

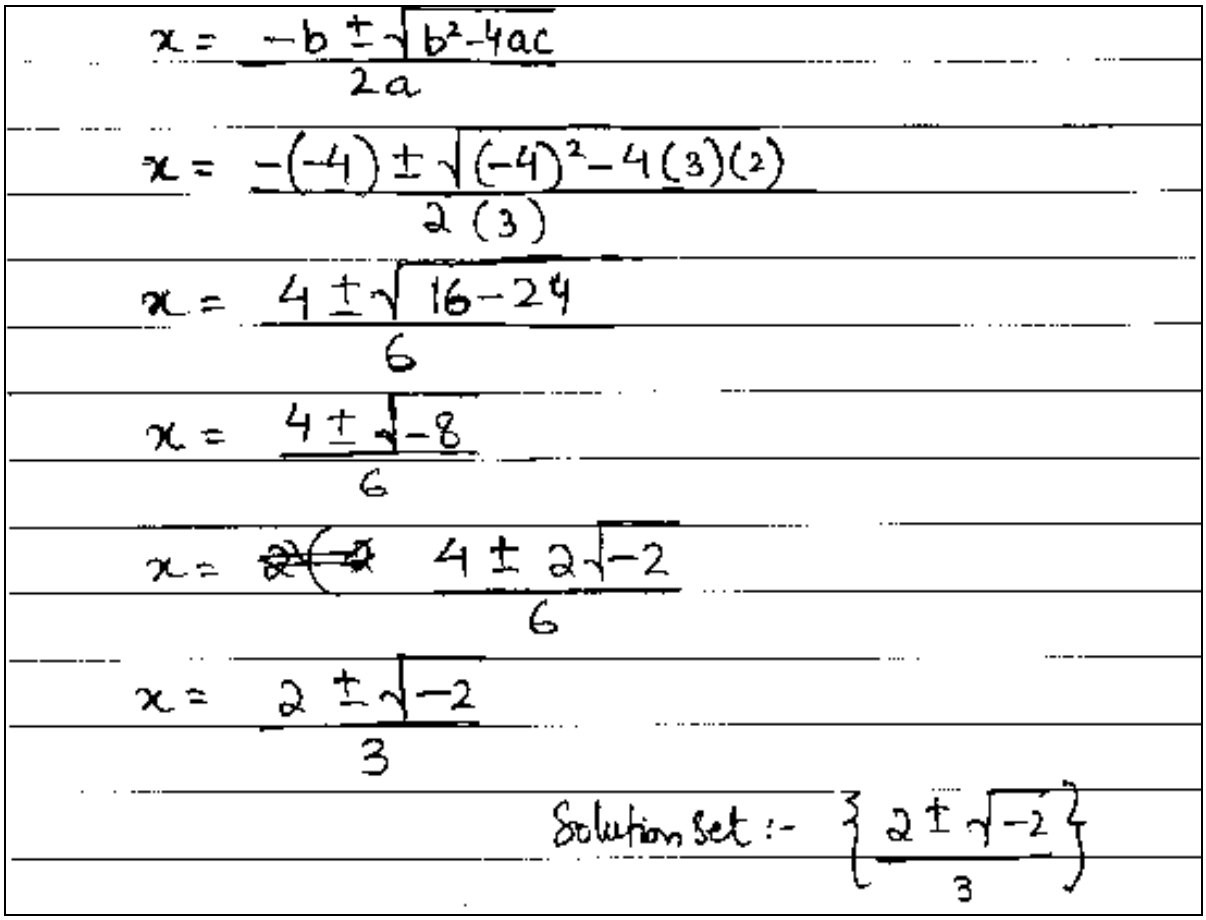
Question 5:

Find the solution set of $3x^2 - 4x + 2 = 0$ by using quadratic formula.

It was generally well-attempted question.

Better responses indicated that candidates were well-versed with the use of quadratic formula. They correctly identified a, *b* and *c* and substituted in the quadratic formula to find the values of *x*.

Example:



Weaker responses indicated that candidates failed to follow the proper steps needed to solve the given quadratic equation. They made mistakes in identifying the values of *a*, *b* and *c* and because of this, they could not reach the required answer

Few responses indicated that candidates made different mistakes in writing formula which are cited below.

$$x = -b \pm \frac{\sqrt{b^2 - 4ac}}{2a}$$

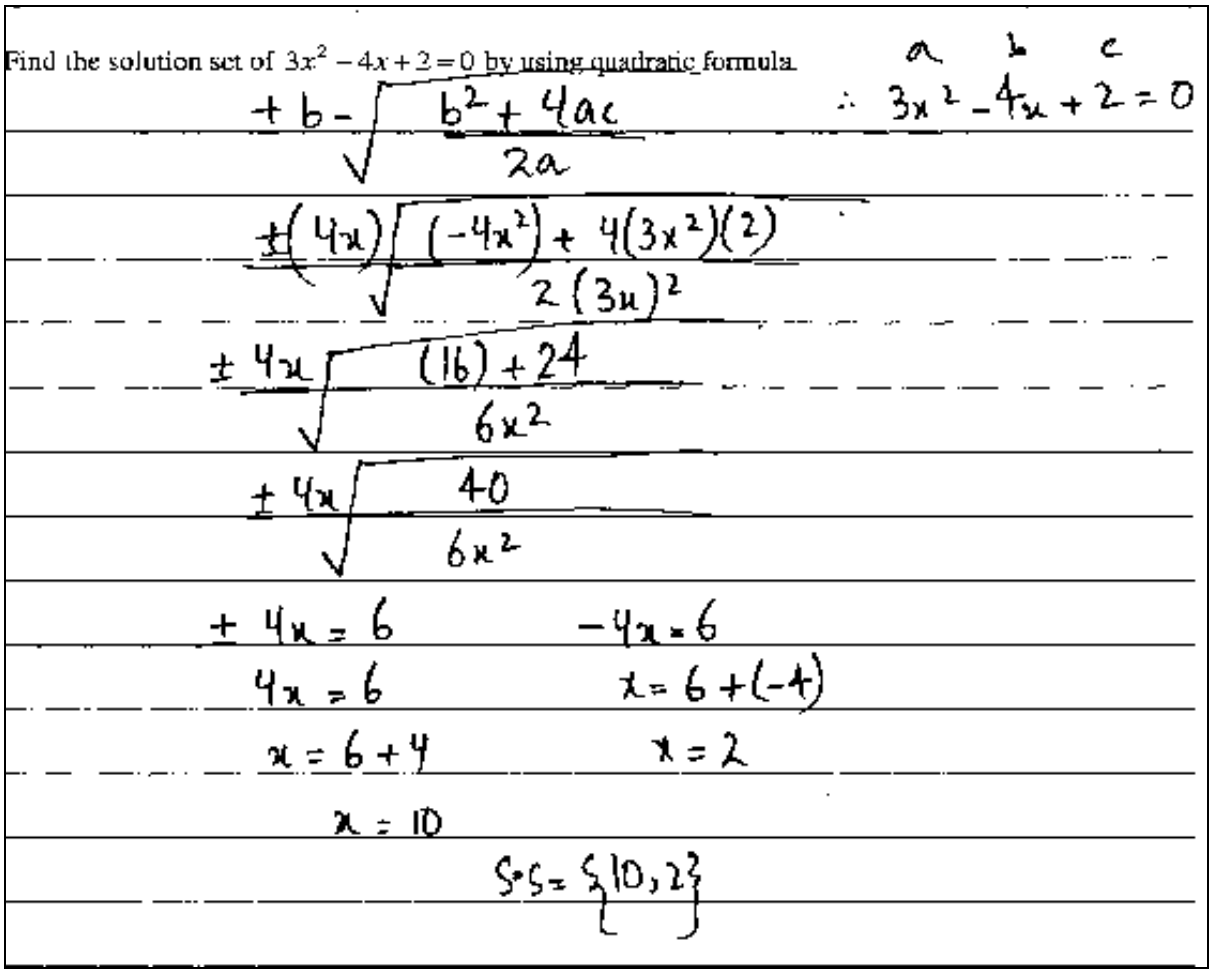
$$x = -b \pm \frac{\sqrt{b^2 + 4ac}}{2a}$$

$$x = \frac{-b\sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{b \pm \sqrt{b^2 - 4ac}}{2a}$$ etc.

Therefore, candidates failed to find the values of *x* for the given quadratic equation.

Example:



Question 6:

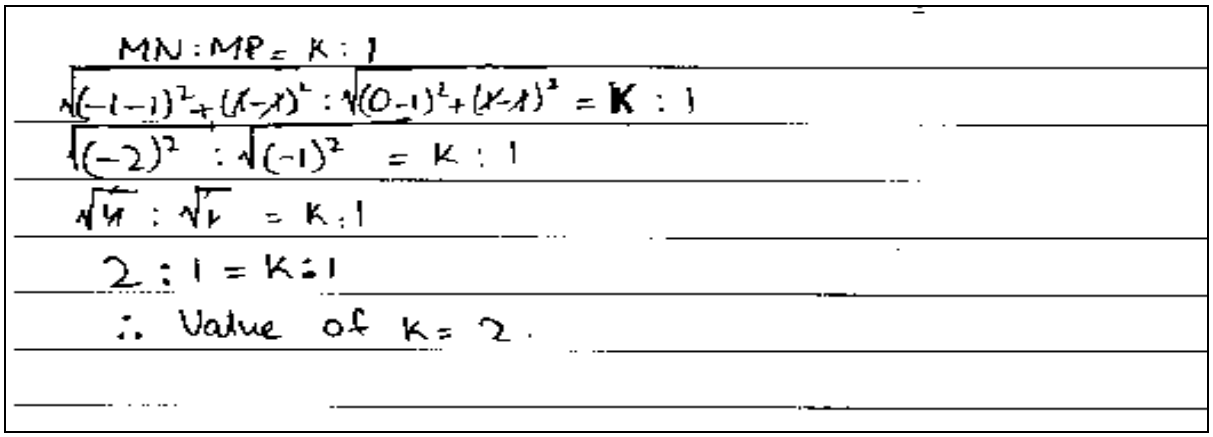
There was a choice between part **a** and part **b** of the question. Both parts were equally attempted by the candidates.

Question 6a:

For the given points *M*, *P* and *N* whose coordinates are $(1,1),(0,1)$ and $(-1,1)$ respectively, find the value of *k* such that $MN:MP=k:1$.

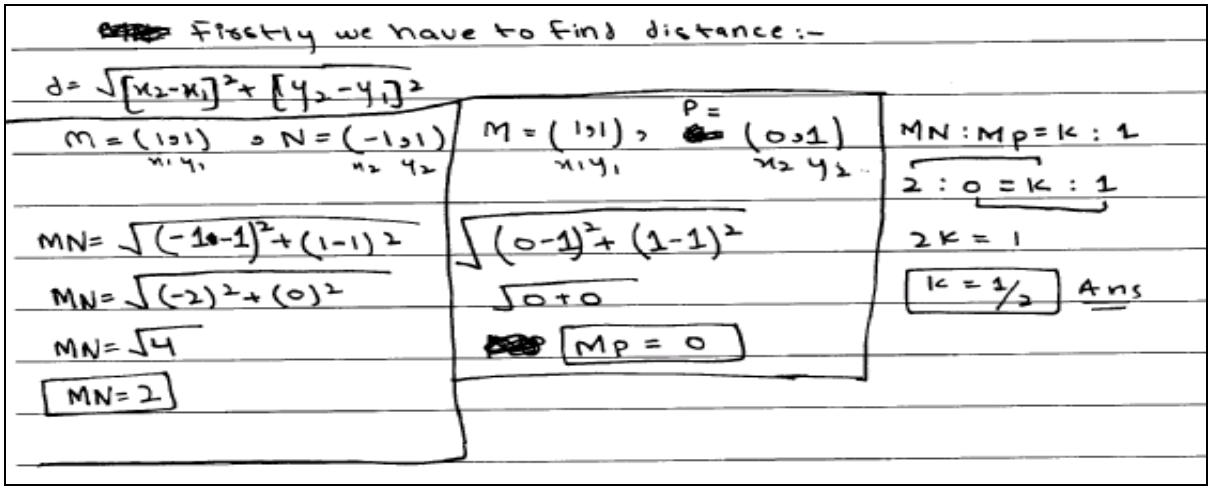
Better responses stated that candidates wrote the correct distance formula, substituted the values of the coordinates correctly and were able to find the value of *k* by using the given condition.

Example:



Weaker responses showed that candidates were unable to comprehend the question which resulted in different types of mistakes especially in distance formula. They made mistakes in substitution of values of the coordinates in the formula. It was also noticed that they made mistakes in simplification after substituting the values in the formula which led to an incorrect value of *k*.

Example:



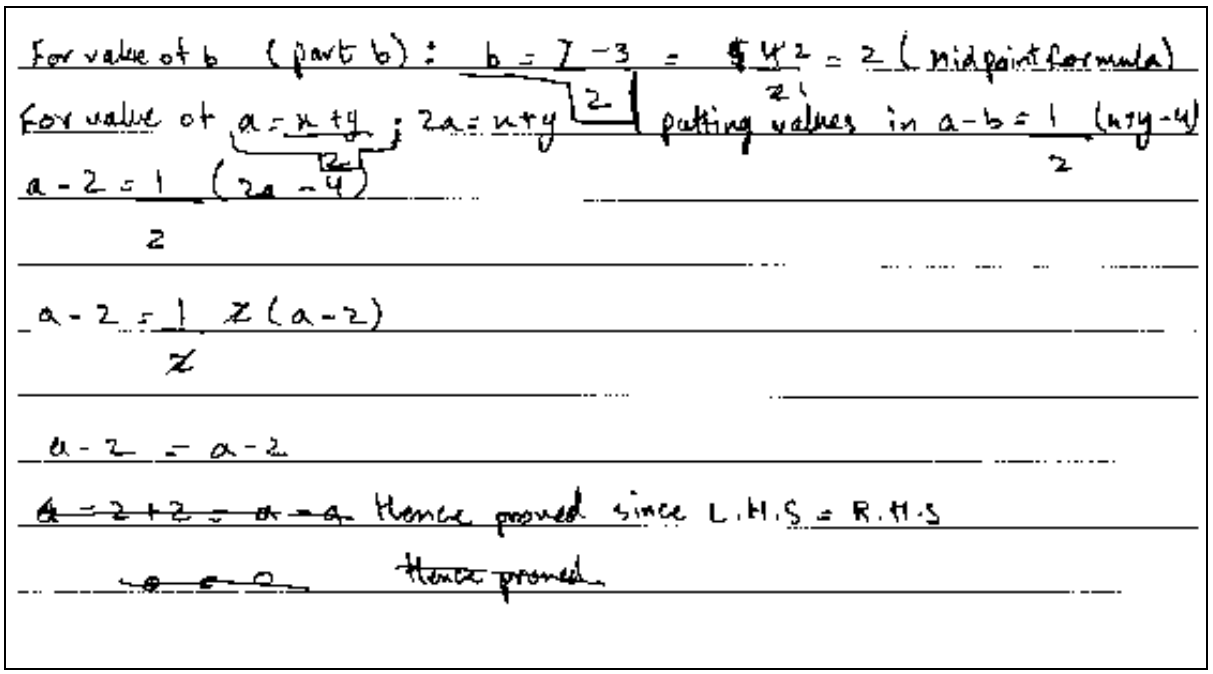
Question 6b:

The midpoint of $P_1(x,-3)$ and $P_2(y,7)$ is (a,b). Show that $a-b=\frac{1}{2}(x+y-4)$.

This question was based on midpoint formula. The candidates who attempted this question generally performed well.

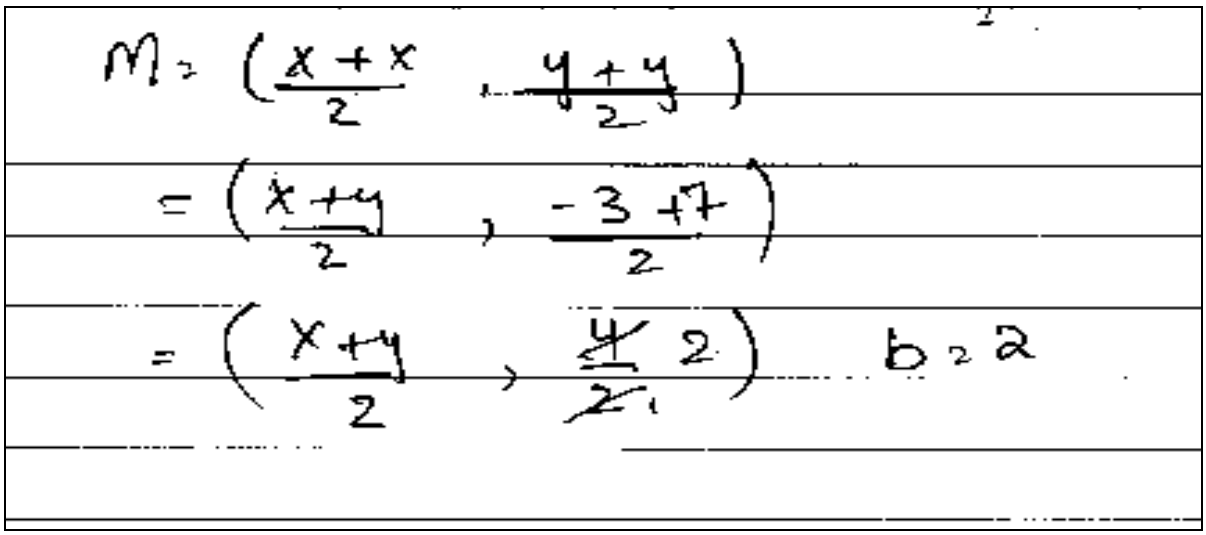
Better responses indicated that candidates comprehended the question well and substituted the values of *a* and *b* correctly in $a-b$. This helped them to reach the required form easily.

Example:



Weaker responses showed that candidates were failed to comprehend the question or applied the formula of midpoint inappropriately. Hence, they failed to go further in reaching the required form. It was also noted that fewer candidates wrote incorrect formula to find the midpoint or applied the distance formula instead of midpoint formula.

Example:



Question 7:

This question offered a choice between part **a** and part **b**. Candidates mostly opted to attempt part **a**.

Question 7a:

A man stands on a flat surface at a distance of 50 metres from a vertical electric pole, *x* metre high. The angle of elevation of the top of electric pole from the man on flat surface is θ.

i. Use the given condition to show that $\theta = \tan^{-1}\left(\frac{x}{50}\right)$

ii. Hence, find the height of electric pole when the angle of elevation is 45°.

iii. After some time, the man moves *y* metres back from his original position. Express in terms of *x* and *y*, the angle of elevation of the top of electric pole from the new position of man.

Better responses revealed that candidates, in part i, applied trigonometric ratio correctly to get $\theta = \tan^{-1}\left(\frac{x}{50}\right)$ and hence, were able to find the height as well. Consequently, they were able to express the angle of elevation quite easily.

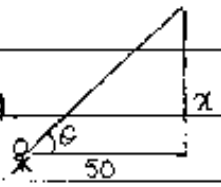
Example:

a. A man stands on a flat surface at a distance of 50 metres from a vertical electric pole, x meter high. The angle of elevation of the top of electric pole from the man on flat surface is θ.

i. Use the given condition to show that $\theta = \tan^{-1}\left(\frac{x}{50}\right)$ (1 Mark)

$\tan\theta = P/B = x/50$

$\theta = \tan^{-1}\left(\frac{x}{50}\right)$ Hence proved.



ii. Hence, find the height of electric pole when the angle of elevation is $45°$. (2 Marks)

$\tan 45° = \frac{x}{50}$

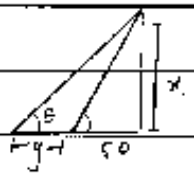
$(1)(50) = x$

$x = 50m$

iii. After some time, the man moves y metres back from his original position. Express in terms of x and y, the angle of elevation of the top of electric pole from the new position of man. (1 Mark)

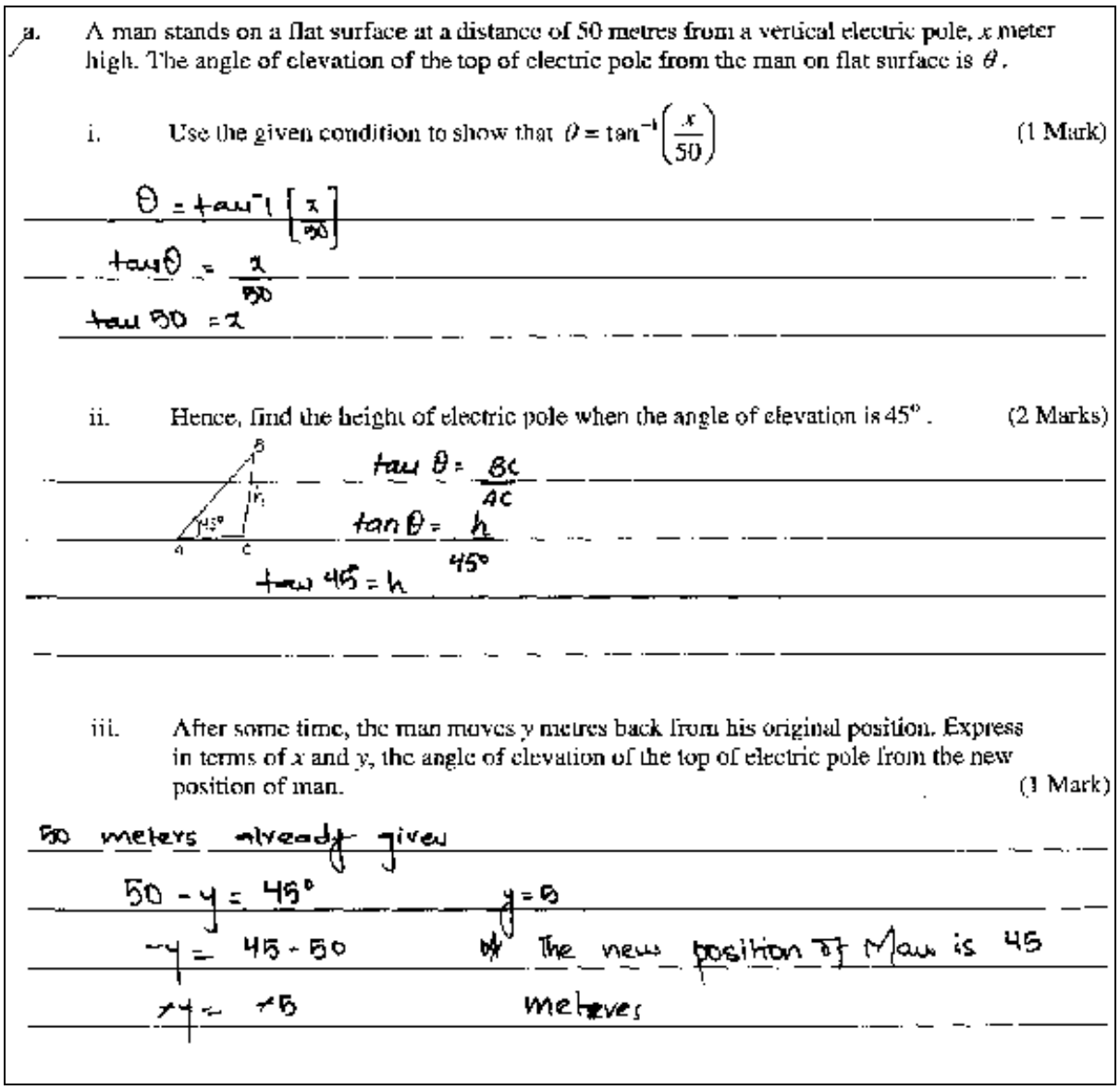
$\tan\theta = \frac{P}{B} = \frac{x}{50+y}$

$\theta = \tan^{-1}\left(\frac{x}{50+y}\right)$



Weaker responses displayed that candidates failed to understand the question and faced difficulty in order to translate mathematically. It was also noted that fewer candidates wrote incorrect value of $\tan 45^{o}$ which led to an incorrect answer. In part iii, some of them wrote $\frac{50}{y+50}$ or just $y+50$ instead of $\frac{x}{y+50}$, which was incorrect.

Example:



a. A man stands on a flat surface at a distance of 50 metres from a vertical electric pole, x meter high. The angle of elevation of the top of electric pole from the man on flat surface is θ.

i. Use the given condition to show that $\theta = \tan^{-1}\left(\frac{x}{50}\right)$ (1 Mark)

$\theta = \tan^{-1}\left[\frac{x}{50}\right]$

$\tan\theta = \frac{x}{50}$

$\tan 50 = x$

ii. Hence, find the height of electric pole when the angle of elevation is 45°. (2 Marks)

$\tan\theta = \frac{BC}{AC}$

$\tan\theta = \frac{h}{45^{\circ}}$

$\tan 45 = h$

iii. After some time, the man moves y metres back from his original position. Express in terms of x and y, the angle of elevation of the top of electric pole from the new position of man. (1 Mark)

50 meters already given

$50 - y = 45^{\circ}$ $\quad y = 5$

$-y = 45 - 50$ ∴ The new position of Man is 45

$-y = -5$ meteres

Better responses reflected that the candidates had good understanding in defining the term radian and well versed with the derivation of $l = r\theta$. In part iii, they were able to manage to get the value of θ with the help of given condition. Generally, It was not a well attempted question.

Example:

i. Define the term radian. (1 Mark)

The angle subtended at the centre of circle by an arc, whose length is equal to the radius of circle, is called radian.

iii. Hence, find θ when $l = \frac{1}{2}r$. (1 Mark)

$\theta = ?$, $l = \frac{1}{2}r$, $r = 2l$

$\Rightarrow l = r\theta \Rightarrow \frac{r}{2} = (2l)(\theta) \Rightarrow \frac{r}{2} = r\left(\frac{1}{r}\right)\theta \Rightarrow \frac{r}{2} = r\theta$

$\Rightarrow \frac{r}{2r} = \theta \Rightarrow \boxed{\theta = \frac{1}{2}}$ ans

Question 7b:

i. Define the term radian.

ii. Derive $l = r\theta$, where r is the radius of the circle, l is the length of circular arc and θ is the central angle measured in radians.

iii. Hence, find θ when $l = \frac{1}{2}r$.

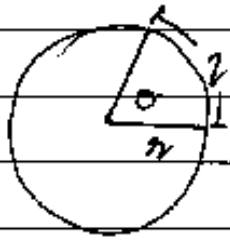
Weaker responses displayed lack of understanding of radians and circular measure and because of this they were unable to derive $l = r\theta$. However, they were able to find θ with the help of given condition but it was also noticed that they could not get the value of θ correctly because of wrong simplification.

Example:

i. Define the term radian. (1 Mark)

It is a term used to measure central angle of the circle.

ii. Derive $l = r\theta$, where r is the radius of the circle, l is the length of circular arc and θ is the central angle measured in radians. (2 Marks)



of circle
length of the arc is directly proportional to its radius & central angle so $l = r\theta$

iii. Hence, find θ when $l = \frac{1}{2}r$. (1 Mark)

$\theta = 1/2$ as $l = r\theta$

Question 8:

The question was generally well-attempted. The question offered a choice between part **a** and part **b**. Candidates mostly attempted part **a** of the question which was based on Pythagorean Theorem and its application.

Question 8a:

Show, by working, that the given lengths $2ab$, $a^2 - b^2$ and $a^2 + b^2$ are the sides of a right angled triangle.

(**Note:** *a* and *b* are positive integers.

Better responses exhibited that candidates comprehended the question well. They correctly applied the Pythagorean Theorem in order to prove that the given lengths are the sides of a right angled triangle.

Example:

$a = a^2 - b^2 \quad b = 2ab \quad c = a^2 + b^2$

$\therefore \quad a^2 + b^2 = c^2$ (Pythagoras theorem)

$(a^2 - b^2)^2 + (2ab)^2 = (a^2 + b^2)^2$

$a^4 - 2a^2b^2 + b^4 + 4a^2b^2 = a^4 + 2a^2b^2 + b^4$

$a^4 + 2a^2b^2 + b^4 = a^4 + 2a^2b^2 + b^4$

OR

$(a^2 + b^2)^2 = (a^2 + b^2)^2$

Hence proved that $(a^2 - b^2)(a^2 + b^2)$ and $2ab$ are the sides of a right angled triangle

Weaker responses indicated that candidates failed to understand the question which led to wrong application of Pythagorean Theorem. They made mistakes in application of theorem, identification of hypotenuse and other sides of the given triangles.

Example:

$H^2 = B^2 + P^2$

$(a^2 + b^2)^2 = (2ab)^2 + (a^2 - b^2)^2$

pythagorus theorem

$(a^2 + b^2)^2 = 4ab^2 + a^4 - 2ab + b^4$

$a^4 + b^4 = a^4 + 2ab + b^4$

$a^4 + b^4 = a^4 + b^4$

Question 8b:

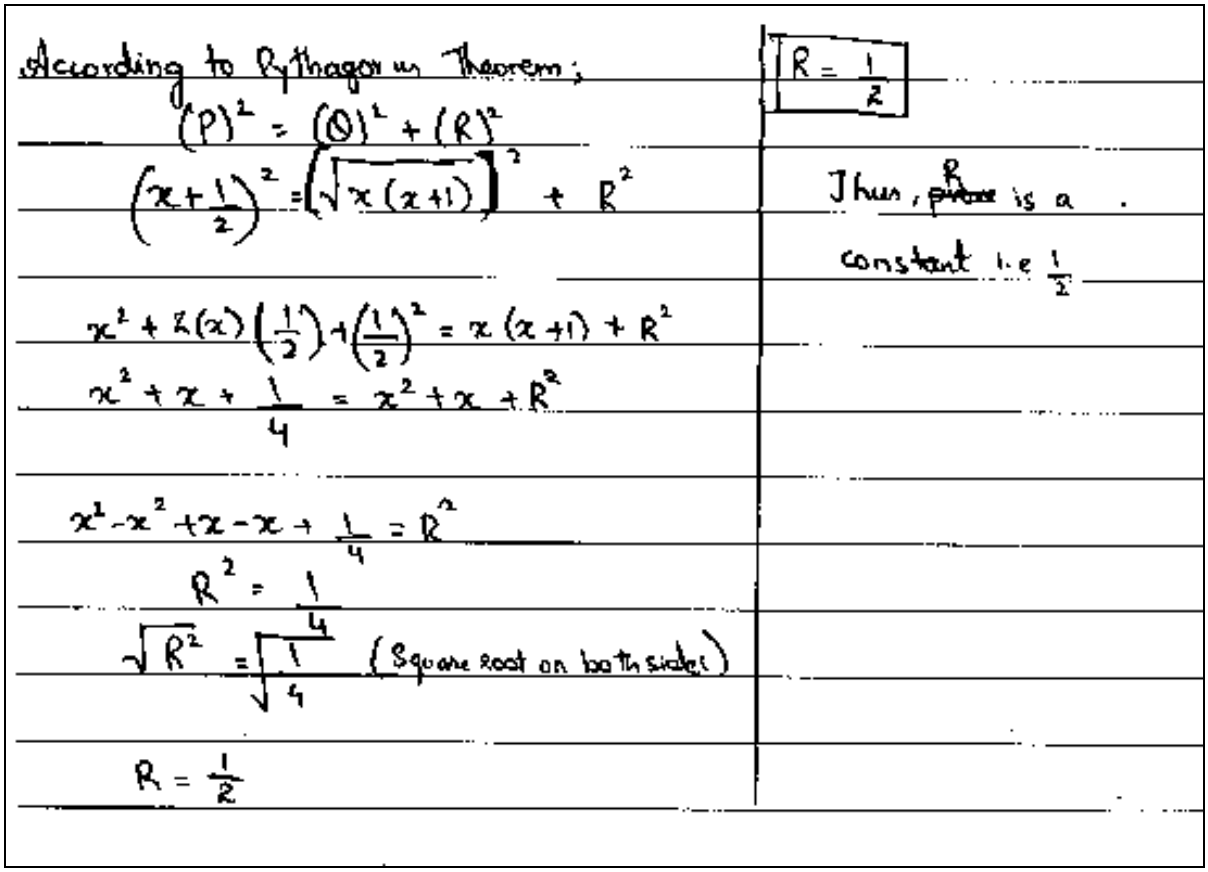
Let *P*, *Q* and *R* be the sides of a right angled triangle such that *P* is the longest side, where $P = x + \frac{1}{2}$ and $Q = \sqrt{x(x+1)}$.

Show that the third side *R* is constant.

As compared to part **a**, this part was attempted by fewer candidates.

Better responses indicated that candidates understood the question well and were able to show the third side is constant by using Pythagoras' theorem correctly. Hence, they showed their command in concept and simplification skills.

Example:



Weaker responses showed that candidates were unable to comprehend the given question and made mistakes in expanding $\left(x+\frac{1}{2}\right)^2$. They did expand $x^2+\frac{1}{4}$ which was incorrect and because of this, candidates could not reach the required result. It was also seen that they were weak in simplifying the algebraic terms and as a result, they lost their marks.

Example:

$P = H^2 = x + \frac{1}{x}$, $Q = \sqrt{x(x+1)}$

(by pythagoras theorem)

$H^2 = B^2 + P^2$

$H^2 = B^2 = P^2$

$\left(x + \frac{1}{2}\right)^2 - \left(\sqrt{x(x+1)}\right)^2 = R^2$

$x^2 + \frac{1}{4} - x(x+1) = R^2$

$x^2 + \frac{1}{4} - x^2 + x = R^2$

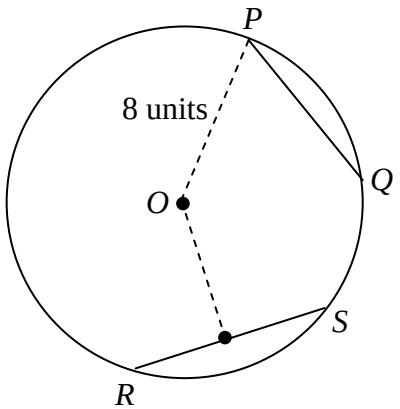
$R^2 = \frac{1}{4} + x$

$R = \frac{1}{2} + \sqrt{x}$

Question 9:

The given circle having centre O has a radius of 8 units. Two equal chords PQ and RS are of length l units respectively.

NOT TO SCALE



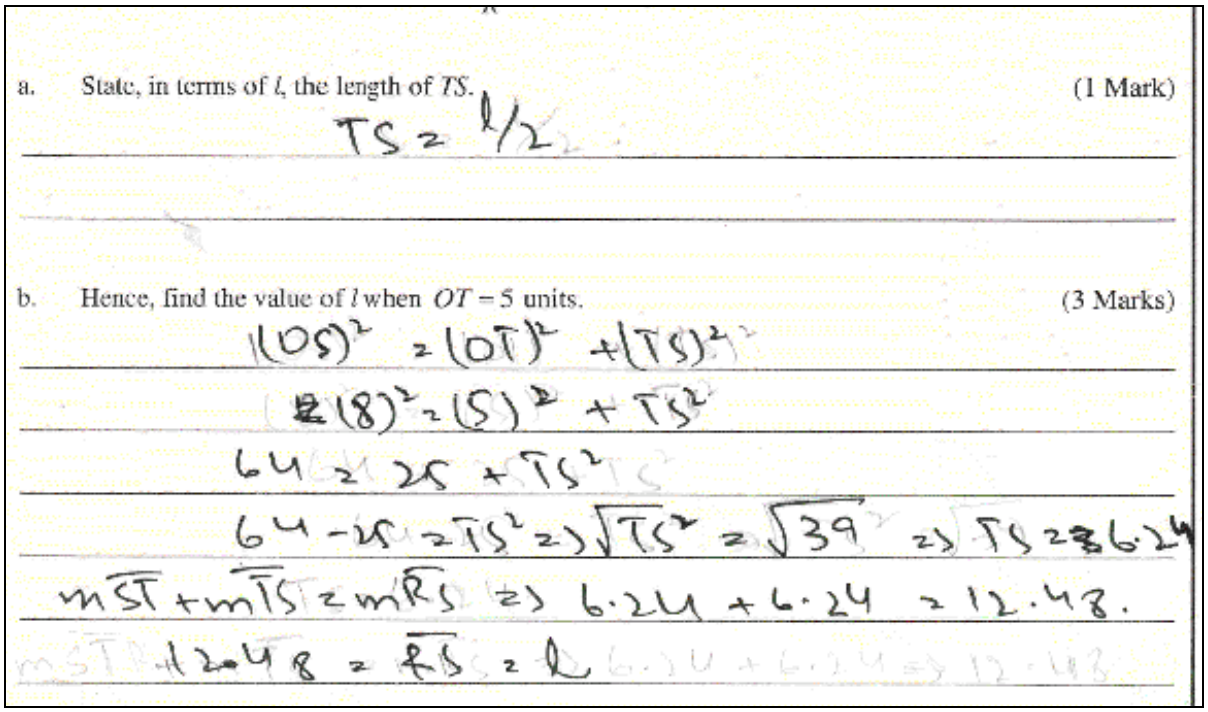
a. State, in terms of l, the length of TS.

b. Hence, find the value of l when $OT = 5$ units.

This question was based on the theorem related to the circle and its properties.

Better responses showed that candidates had command over the theorems and their application. Candidates were able to find the required length in terms of *l*. They did well to find the value of *l* by applying Pythagoras' theorem.

Example:



Weaker responses indicated that the candidates failed to comprehend the properties of circles and therefore, were unable to find the required lengths correctly. They made mistakes in Pythagoras' theorem as they wrote $l^2 = 8^2 - 5^2$ instead of $\left(\frac{l}{2}\right)^2 = 8^2 - 5^2$ which was actually incorrect. Because of this, they were unable to get the desired answer which resulted in the loss of marks.

Example:

a. State, in terms of l, the length of TS. (1 Mark)

$TS = \frac{l}{2}$ units (A straight line drawn from the centre of circle on a chord, bisects it).

b. Hence, find the value of l when $OT = 5$ units. (3 Marks)

In $\triangle OTR$, $OR = 8$ units (radius of circle), $OT = 5$ units (given)

while $RT = RS/2 = l/2$

Using pythagoras theorem,

$(OR)^2 = (OT)^2 + (RT)^2$

$8^2 = 5^2 + (l-2)^2$

$64 = 25 + (l-2)^2$

$(l-2)^2 = 64 - 25$

$(l-2)^2 = 39$

root both sides

$\sqrt{(l-2)^2} = \sqrt{39}$

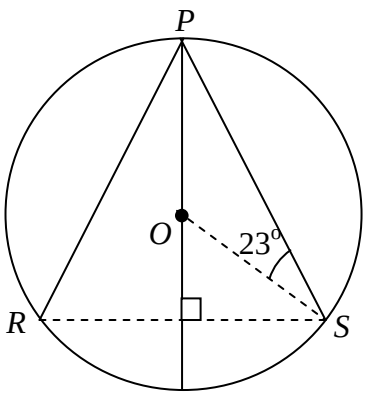
$l - 2 = 6.244$

$l = 6.244 + 2 = 8.244$

~~$l = 8.244$~~

Question 10:

In the diagram, O is the centre of the circle and it is given that $\angle OSP = 23^o$ and $PS = PR$.

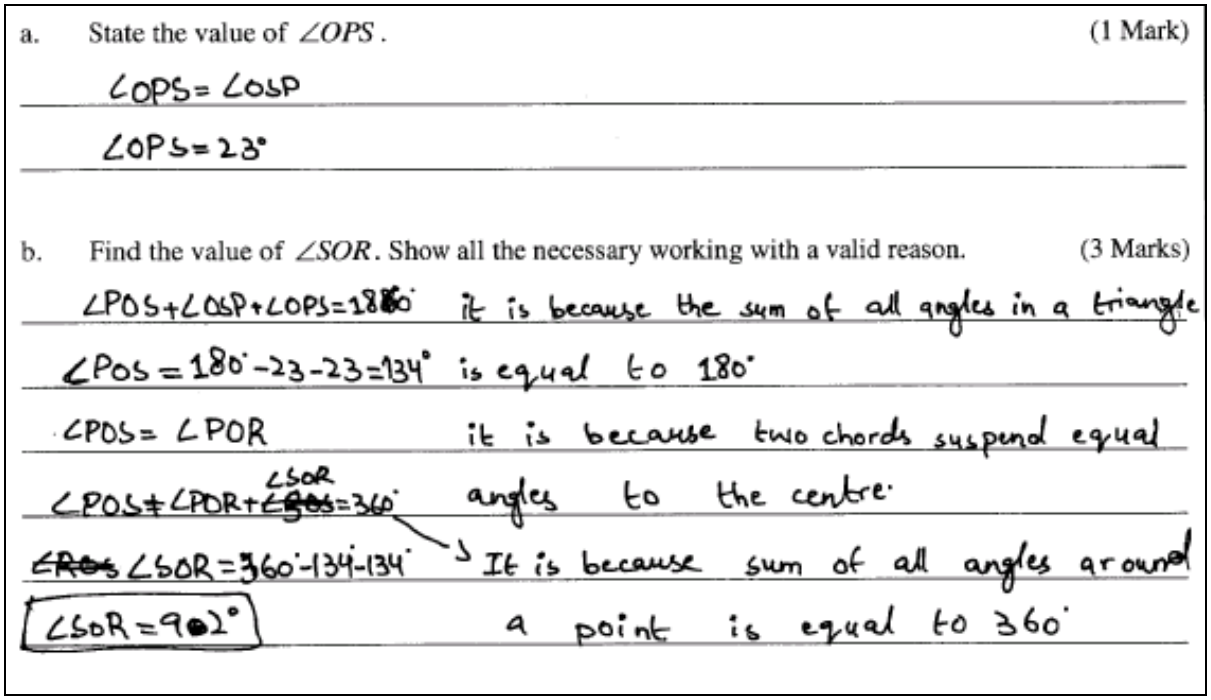


a. State the value of $\angle OPS$.

b. Find the value of $\angle SOR$. Show all the necessary working with a valid reason.

Better responses displayed that candidates had command over the theorems and applied relevant theorems to find $\angle OPS$ and $\angle SOR$ correctly. This showed candidates' ability and hence, they were able to give valid reason for their answers.

Example:


a. State the value of $\angle OPS$. (1 Mark)

$\angle OPS = \angle OSP$

$\angle OPS = 23°$

b. Find the value of $\angle SOR$. Show all the necessary working with a valid reason. (3 Marks)

$\angle POS + \angle OSP + \angle OPS = 180°$ it is because the sum of all angles in a triangle

$\angle POS = 180° - 23 - 23 = 134°$ is equal to 180°

$\angle POS = \angle POR$ it is because two chords suspend equal

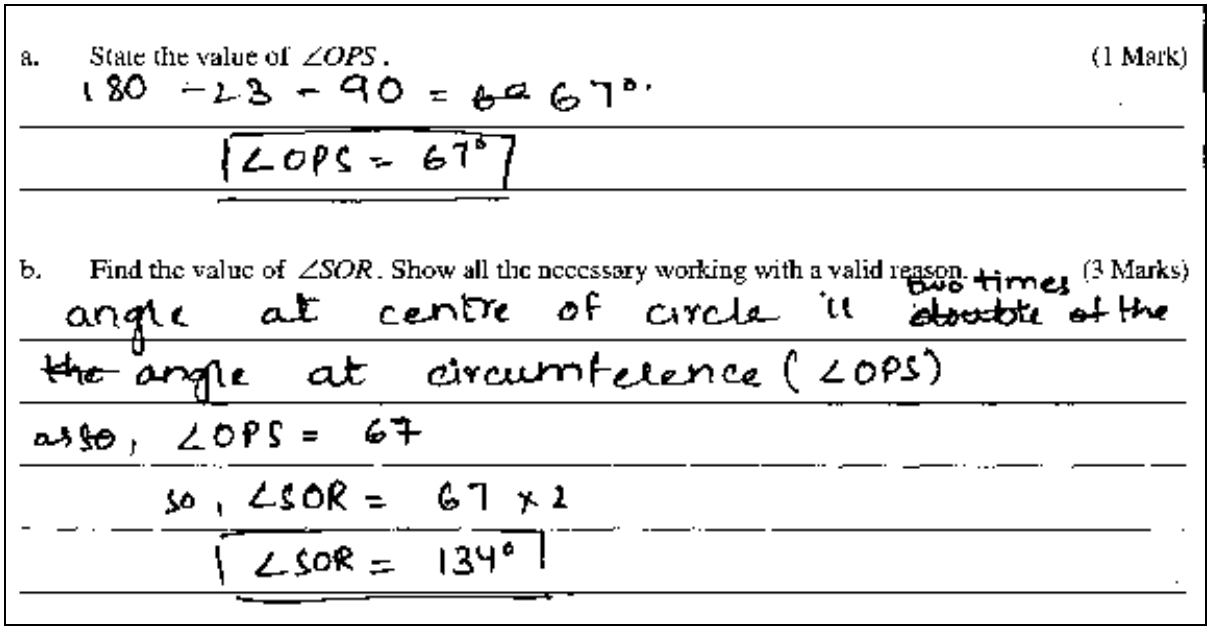
$\angle POS + \angle POR + \angle SOR = 360°$ angles to the centre.

$\angle SOR = 360° - 134 - 134$ It is because sum of all angles around

$\angle SOR = 92°$ a point is equal to 360°

Weaker responses indicated that the candidates were unable to comprehend the theorem to be applied in the given question. Hence, they failed to fulfill the requirement of the question and found it difficult to find the required angle. It was also seen that candidates were confused in giving a reason to justify their answers.

Example:


a. State the value of $\angle OPS$. (1 Mark)

$180 - 23 - 90 = 67°$

$\angle OPS = 67°$

b. Find the value of $\angle SOR$. Show all the necessary working with a valid reason. (3 Marks)

angle at centre of circle is two times the angle at circumference ($\angle OPS$)

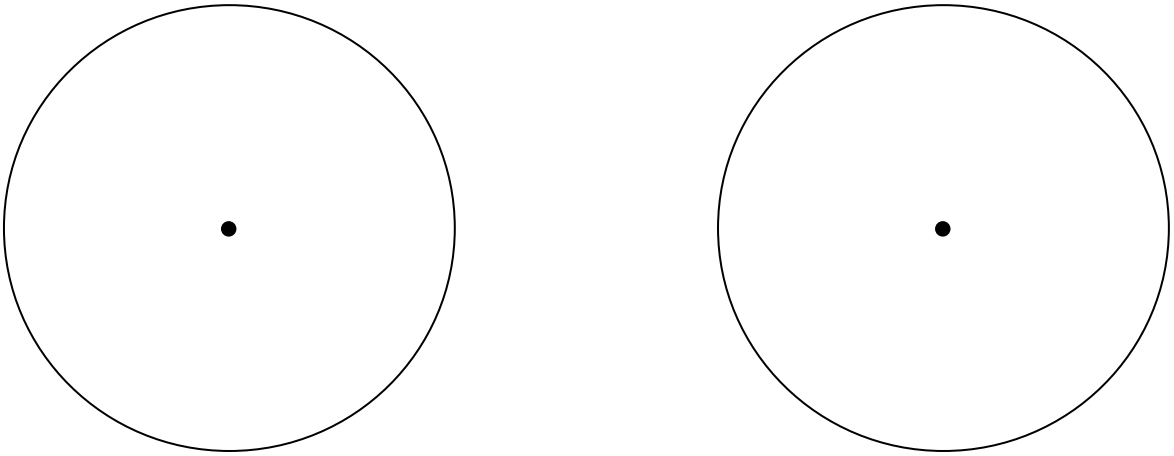
also, $\angle OPS = 67$

so, $\angle SOR = 67 \times 2$

$\angle SOR = 134°$

Question 11:

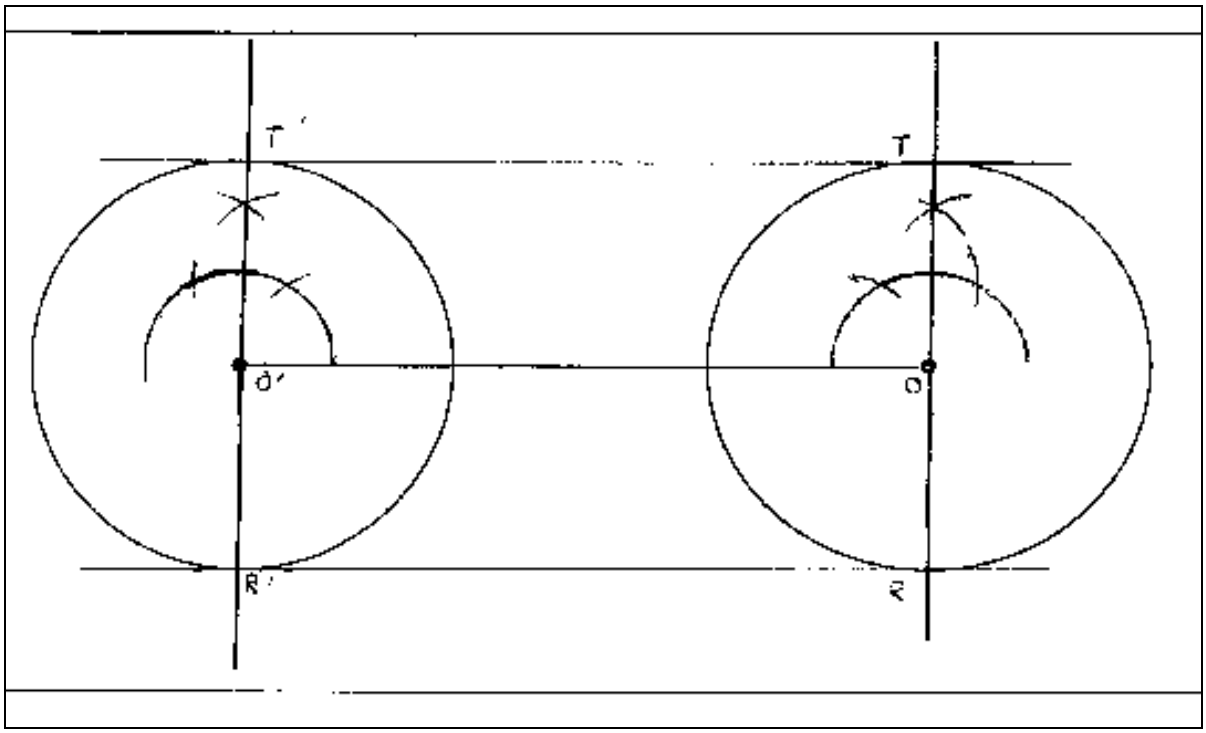
Two circles of same radii with their centres are shown in the given diagram. Draw direct common tangents to the given circles. Show all the necessary steps.



This question was based on direct common tangents .

Better responses displayed that candidates had good command over geometrical construction and had clear understanding of the process of drawing direct common tangents. They first joined the centres and drew the perpendicular line from centres of both the circles to the circumferences of the circle and finally, drew the direct common tangents.

Example:



Weaker responses indicated that candidates had lack of understanding and failed to draw the direct common tangents. A few candidates drew perpendicular lines only by using their judgment instead of following proper steps required to draw the direct common tangents. Because of this, they losen their marks.

Example:

